

Pattern Printting HomeWork

1. Star Rectangle

```
#include <iostream>
using namespace std;

int main(){
    int m,n;
    cout <<"Number of Rows: ";
    cin >>m;
    cout <<"Number of Cols: ";
    cin >>n;

    for(int i = 1;i <= m;i++){
        for(int j = 1; j<=n;j++){
            cout<<"* ";
        }
        cout <<endl;
    }
}
```

Output

```
Number of Rows: 5
Number of Cols: 6
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

2.Star Square

```
#include <iostream>
using namespace std;

int main(){
    int n;
    cout <<"Number of Star: ";
    cin >>n;

    for(int i = 1;i <= n;i++){
        for(int j = 1;j <= n;j++){
            cout <<"* ";
        }
    }
}
```

```
    }  
    cout <<endl;  
}  
}
```

OutPut

```
Number of Star: 5  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

3.Number Square

```
#include <iostream>  
using namespace std;  
  
int main(){  
  
    int num;  
    cout <<"Number of Number: ";  
    cin >>num;  
  
    for(int i = 1;i <= num;i++){  
        for(int j = 1;j <= num;j++){  
            cout <<j <<" ";  
        }  
        cout <<endl;  
    }  
  
}
```

Output (Number Square)

```
Number of Number: 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

4.Number Square 2

```
#include <iostream>
using namespace std;

int main(){

    int num;
    cout <<"Number of Number: ";
    cin >>num;

    for(int i = 1;i <= num;i++){
        for(int j = 1;j <= num;j++){
            cout <<i <<" ";
        }
        cout <<endl;
    }
}
```

Output (Number Square 2)

```
Number of Number: 5
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
```

5.Alphabet Square

```
#include <iostream>
using namespace std;

int main() {
```

```
int n;
cout << "Number of alphabet: ";
cin >> n;

for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        cout << (char)(j + 64) << " ";
    }
    cout << endl;
}
```

Output (Alphabet Square)

```
Number of alphabet: 4
A B C D
A B C D
A B C D
A B C D
```

6.alphbet Square 2

```
#include <iostream>
using namespace std;

int main() {

    int n;
    cout << "Number of alphabet: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            cout << (char)(j + 96) << " ";
        }
        cout << endl;
    }
}
```

Output(alphbet Square 2)

```
Number of alphabet: 5
a b c d e
a b c d e
a b c d e
```

```
a b c d e
a b c d e
```

7.Alphabet Square 3

```
#include <iostream>
using namespace std;

int main() {

    int n;
    cout << "Number of alphabet: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            cout << (char)(i + 64) << " ";
        }
        cout << endl;
    }

}
```

Output(Alphabet Square 3)

```
Number of alphabet: 4
A A A A
B B B B
C C C C
D D D D
```

8.Alphabet Square 4

```
#include <iostream>
using namespace std;

int main() {

    int n;
    cout << "Number of alphabet: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
```

```
        if(i % 2 == 0){
            cout << (char)(i + 64) << " ";
        }else{
            cout << (char)(i + 96) << " ";
        }
    }
    cout << endl;
}
}
```

Output(Alphabet Square 4)

```
Number of alphabet: 4
a a a a
B B B B
c c c c
D D D D
```

9.Star Triangle

```
#include <iostream>
using namespace std;

int main(){
    int n;
    cout <<"Num of star: ";
    cin >>n;

    for(int i = 1; i <= n; i++){
        for(int j = 1; j <= i; j++){
            cout <<"* ";
        }
        cout <<endl;
    }
}
```

Output(Star Triangle)

```
Num of star: 5
*
* *
* * *
* * * *
* * * * *
```

```
* * * * *
```

10.Number Triangle

```
#include <iostream>
using namespace std;

int main(){
    int n;
    cout <<"Num of nums: ";
    cin >>n;

    for(int i = 1; i <= n; i++){
        for(int j = 1; j <= i; j++){
            cout <<j <<" ";
        }
        cout <<endl;
    }
}
```

Output(Number Triangle)

```
Num of nums: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

11.Number Triangle 2

```
#include <iostream>
using namespace std;

int main(){
    int n;
    cout <<"Num of nums: ";
    cin >>n;

    for(int i = 1; i <= n; i++){
        for(int j = 1; j <= i; j++){
            cout <<i <<" ";
        }
        cout <<endl;
    }
}
```

```
}  
}
```

Output(Number Triangle 2)

```
Num of nums: 6  
1  
2 2  
3 3 3  
4 4 4 4  
5 5 5 5 5  
6 6 6 6 6 6
```

12.Alphabet Triangle 2

```
#include <iostream>  
using namespace std;  
  
int main(){  
    int n;  
    cout <<"Num of alphabet: ";  
    cin >>n;  
  
    for(int i = 1; i <= n; i++){  
        for(int j = 1; j <= i; j++){  
            cout << (char)(j+64) <<" ";  
        }  
        cout <<endl;  
    }  
}
```

Output(Alphabet Triangle)

```
Num of alphabet: 5  
A  
A B  
A B C  
A B C D  
A B C D E
```

13.Alphabet Triangle 2


```
#include <iostream>
using namespace std;

int main(){
    int n;
    cout <<"Num of alphabet: ";
    cin >>n;

    for(int i = 1; i <= n; i++){
        for(int j = 1; j <= i; j++){
            cout << (char)(i+64) <<" ";
        }
        cout <<endl;
    }
}
```

Output(Alphabet Triangle 2)

```
Num of alphabet: 5
A
B B
C C C
D D D D
E E E E E
```

14.Alpha Numeric Triangle

```
#include <iostream>
using namespace std;

int main(){
    int n;
    cout <<"Num of alphabet: ";
    cin >>n;

    for(int i = 1; i <= n; i++){
        for(int j = 1; j <= i; j++){
            if(i % 2 == 0){
                cout << (char)(j+64) <<" ";
            }else{
                cout << j <<" ";
            }
        }
        cout <<endl;
    }
}
```

```
}
```

Output(Alpha Numeric Triangle)

```
Num of alphabet: 5
1
A B
1 2 3
A B C D
1 2 3 4 5
```

15.Star Triangle Horizontal Flipped

```
#include <iostream>
using namespace std;

int main(){
    int s;
    cout <<"Number of Star: ";
    cin >>s;

    for(int i = 1;i <= s;i++){
        for(int j = 1; j <= s + 1 - i;j++){
            cout <<"* ";
        }
        cout <<endl;
    }
}
```

Output()

```
Number of Star: 5
* * * * *
* * * *
* * *
* *
*
```

16.Flipped Number Triangle

```
#include <iostream>
using namespace std;

int main(){
    int s;
    cout <<"Number of nums: ";
    cin >>s;

    for(int i = 1;i <= s;i++){
        for(int j = 1; j <= s + 1 - i;j++){
            cout <<j <<" ";
        }
        cout <<endl;
    }
}
```

Output(Flipped Number Triangle)

```
Number of nums: 5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

17.Flipped Number Triangle 2

```
#include <iostream>
using namespace std;

int main(){
    int s;
    cout <<"Number of Nums: ";
    cin >>s;

    for(int i = 1;i <= s;i++){
        for(int j = 1; j <= s + 1 - i;j++){
            cout <<i <<" ";
        }
        cout <<endl;
    }
}
```

Output(Flipped Number Triangle 2)

```
Number of Nums: 5
1 1 1 1 1
2 2 2 2
3 3 3
4 4
5
```

18.Flipped Alphabet Triangle

```
#include <iostream>
using namespace std;

int main(){
    int s;
    cout <<"Number of Alphabet: ";
    cin >>s;

    for(int i = 1;i <= s;i++){
        for(int j = 1; j <= s + 1 - i;j++){
            cout <<(char)(j + 64) <<" ";
        }
        cout <<endl;
    }
}
```

Output(Flipped Alphabet Triangle)

```
Number of Alphabet: 5
A B C D E
A B C D
A B C
A B
A
```

19.Flipped Alphabet Triangle 2

```
#include <iostream>
using namespace std;

int main(){
    int s;
    cout <<"Number of Star: ";
```

```

cin >>s;

for(int i = 1;i <= s;i++){
    for(int j = 1; j <= s + 1 - i;j++){
        cout <<(char) (i + 64)<<" ";
    }
    cout <<endl;
}
}

```

Output(Flipped Alphabet Triangle 2)

```

Number of Star: 5
A A A A A
B B B B
C C C
D D
E

```

20.Hollow Star Rectangle

```

#include <iostream>
using namespace std;

int main() {
    int m, n;

    cout << "Number of Rows: ";
    cin >> m;

    cout << "Number of Cols: ";
    cin >> n;

    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {

            if (i == 1 || i == m || j == 1 || j == n) {
                cout << "*" << " ";
            } else {
                cout << "  ";
            }
        }
        cout << endl;
    }
}

```

Output(Hollow Star Rectangle)

```

Number of Rows: 5
Number of Cols: 6
* * * * *
*           *
*           *
*           *
* * * * *

```

21.Star Plus

```

#include <iostream>
using namespace std;

int main() {
    int o;
    cout << "Enter only Odd Nums: ";
    cin >> o;

    int mid = o/2+1;

    for (int i = 1; i <= o; i++) {
        for (int j = 1; j <= o; j++) {
            if(i==mid || j==mid){
                cout <<"*" <<" ";
            }else{
                cout <<"  ";
            }
        }
        cout << endl;
    }
}

```

Output(Star Plus)

```

Enter only Odd Nums: 5
    *
    *
* * * * *
    *
    *

```

22.Star Cross

```

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter only Odd Nums: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if(i == j || i + j == n+1){
                cout <<"* ";
            }else{
                cout <<" ";
            }
        }
        cout << endl;
    }
}

```

Output(Star Cross)

```

Enter only Odd Nums: 5
*       *
 *     *
  *   *
   * *
  * *
 *   *
*       *

```

23.Floyd's Triangle

```

#include <iostream>
using namespace std;

int main(){
    int num;
    cout <<"Enter no. of rows: ";
    cin >>num;

    int a = 1;
    for(int i = 1; i <= num; i++){
        for (int j = 1; j <=i;j++){
            cout <<a++<<" ";
        }
        cout <<endl;
    }
}

```

```
}
```

Output(Floyd's Triangle)

```
Enter no. of rows: 5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

24.Odd Number Triangle

```
#include <iostream>
using namespace std;

int main(){
    int num;
    cout <<"Enter no. of rows: ";
    cin >>num;

    int a = 1;
    for(int i = 1; i <= num; i++){
        for (int j = 1; j <=i;j++){
            cout <<2 * j -1 <<" ";
        }
        cout <<endl;
    }
}
```

Output(Odd Number Triangle)

```
Enter no. of rows: 5
1
1 3
1 3 5
1 3 5 7
1 3 5 7 9
```